Hema Sundhar Rayapureddy

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PROFESSIONAL SUMMARY

Second-year B.Tech student in Mathematics and Computing at RGIPT with a strong focus on Machine Learning and Data Science. Proficient in Python and SQL with hands-on experience building end-to-end machine learning systems, including data preprocessing, feature engineering, model development, and deployment.Developed a production-grade credit risk prediction system using 2.2M+ real-world financial records, implementing advanced models (XGBoost) and model explainability techniques (SHAP) for interpretable decision-making. Experienced in REST API development using FastAPI, data analysis, and scalable system design.Strong foundation in data structures, algorithms, and mathematical modeling, with a keen interest in applying data-driven solutions to real-world problems in fintech and AI.

EDUCATION

B.Tech – Mathematics and Computing Rajiv Gandhi Institute of Petroleum Technology (RGIPT) 2024 – 2028

Senior Secondary (Class XII) Tirumala Educational Institutes, Rajamahendravaram CGPA: 9.95

Secondary School (Class X) LOYOLA (EM) School, Kothapet CGPA: 8.13

PROJECTS

Credit Risk & Loan Default Prediction System (Production-Grade ML + Real-Time API + Explainable AI)

Dec 2025 – Apr 2026

- Built a production-grade end-to-end Machine Learning system to predict loan default risk using 2.2M+ real-world financial records
- Developed and optimized models (XGBoost, Random Forest, Logistic Regression) achieving ROC-AUC of 0.74 with imbalanced data handling
- Engineered advanced financial features (debt-to-income ratio, credit utilization, inquiry patterns) to improve model performance
- Implemented model explainability using SHAP for interpretable risk predictions and regulatory compliance
- Designed and deployed REST APIs using FastAPI enabling real-time prediction (<100ms latency)
- Built an interactive dashboard using Streamlit for risk visualization, decision support, and user input simulation
- Designed scalable system architecture integrating ML pipeline, API layer, and database with Docker-based deployment
- Developed a risk scoring system (0–100) with business rules for automated loan approval, rejection, and manual review
- Processed 2.2M+ financial records achieving real-time prediction with <100ms latency
- Implemented end-to-end ML pipeline with real-time inference, model monitoring, and scalable API integration simulating real-world fintech systems
- **Live Project: https://huggingface.co/spaces/RayapureddyHemaSundhar/credit-risk-system**

E-Commerce Analytics SQL Platform

Jan 2026 – Mar 2026

- Designed a production-ready relational database system for e-commerce analytics

- Created 12 normalized tables for customers, orders, products, inventory, and suppliers
- Implemented advanced SQL analytics including RFM segmentation, cohort analysis, and market basket analysis
- Built triggers and stored procedures for automated calculations and inventory alerts
- Developed analytical views and queries for business intelligence insights

MATLAB Data Preprocessing & Regression Pipeline

Jan 2026 – Mar 2026

- Built a modular MATLAB pipeline for preprocessing tabular and time-series datasets
- Implemented missing value handling, outlier detection, normalization, and feature engineering
- Applied feature selection methods including PCA, correlation filtering, and RFE
- Trained and evaluated regression models with cross-validation and Bayesian hyperparameter tuning
- Generated automated HTML reports for analysis

Python Data Structures & Algorithms Toolkit

Dec 2025 – Mar 2026

- Implemented fundamental data structures including stacks, queues, linked lists, trees, and graphs
- Developed sorting algorithms such as Quick Sort and Merge Sort from scratch
- Analyzed time and space complexity for each algorithm
- Built modular Python classes and tested using interactive notebooks

Customer Churn Prediction System

Dec 2025 – Mar 2026

- Developed a machine learning model to predict customer churn using behavioral and account data
- Performed data preprocessing, feature engineering, and exploratory data analysis
- Trained models including Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting
- Achieved ~75% accuracy and identified key churn factors

TECHNICAL SKILLS

Python	Scikit-learn	Model Evaluation (ROC-AUC, Precision-Recall)	Pandas	Real-time Inference	Streamlit
C++	XGBoost	Imbalanced Learning	NumPy	FastAPI	Plotly
SQL	Feature Engineering	Model Explainability (SHAP)	Matplotlib	REST API Development	MySQL
EDA	PostgreSQL	SQLAlchemy	Docker	API Integration	Seaborn
Git	Jupyter Notebook	Model Deployment	VS Code		

CORE ML CONCEPTS

Supervised Learning, Classification, Feature Engineering, Model Evaluation, Bias-Variance Tradeoff, Imbalanced Data Handling

RELEVANT COURSEWORK

Machine Learning, Probability & Statistics, Linear Algebra, Data Structures & Algorithms